

Selection Shapes the Boundary: A Preregistered Replication of Monotonicity and Label Agreement in Unselected NLI Populations

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Abstract

Prior work on human label variation (HLV) in natural language inference (NLI) has often relied on re-annotation resources that select items by disagreement level. An earlier study (Choi, 2026) found that hypotheses containing non-upward monotonicity operators showed lower label agreement in ChaosNLI (Cliff’s $\delta = -0.284$), which is restricted to items whose majority label carries exactly three of five votes. We preregistered a replication of this boundary in the unselected populations that ChaosNLI was drawn from: the SNLI and MultiNLI development sets, using the same operator tagger and a four-level ordinal agreement outcome. The registered prediction fails. All seven contrasts return a positive Cliff’s δ (non-upward items agree slightly more, not less), the only significant confirmatory contrast has the opposite sign to the registration, and every effect is far below our smallest effect size of interest (0.10). Robustness checks support the measurement: simulated tagger misclassification shrinks the effects rather than manufacturing them, and a manual re-tagging audit reaches four-class agreement of 0.875 on a fresh 200-item sample. We conclude that the earlier negative boundary is plausibly a structure conditional on low-agreement selection rather than a population-level property, and that HLV structure claims built on selected re-annotation resources should state their selection conditional explicitly.

1 Introduction

Human label variation in NLI is increasingly treated as signal rather than annotation noise: multiple labels per item reflect genuine interpretive variation that models and evaluations should represent (Plank, 2022; Pavlick and Kwiatkowski, 2019). Under this perspectivist frame, disagreement is a measurement target, and a natural research question is whether linguistic properties of an item predict how much annotators will disagree on it.

One candidate predictor is monotonicity. Choi (2026) tagged hypothesis sentences for monotonicity operators and found that in ChaosNLI, items whose hypotheses contained non-upward operators (downward-entailing or non-monotone triggers) had lower label agreement than purely upward items, with a Cliff’s δ of -0.284 (95% CI $[-0.332, -0.235]$) on a 100-label entropy outcome.

That estimate, however, was conditional on a strong selection rule. ChaosNLI re-annotated only items whose majority label carried exactly three of the five original SNLI/MNLI annotator labels. Whether the monotonicity boundary exists in the unselected populations those items came from was never measured. This paper measures it, under pre-registration. We apply the identical tagger, whose rules were frozen before analysis, to the SNLI and MNLI development sets and test the registered prediction that non-upward items sit lower on a four-level agreement ordinal built from the original five labels.

The registered prediction is rejected. The sign reverses in all seven contrasts, and every effect is small: the largest δ is $+0.059$, far below our pre-registered smallest effect size of interest (SESOI) of 0.10, and no confidence interval reaches it. The contribution of this paper is the measured selection dependence itself: a boundary that is present inside a disagreement-selected resource can vanish and even faintly reverse in the unselected population.

2 Related Work

NLI disagreement resources. SNLI and MNLI development items carry five labels each, the original writer’s label plus four validation labels (Bowman et al., 2015; Williams et al., 2018); ChaosNLI re-annotates a selected subset with 100 labels each (Nie et al., 2020, Section 3.1). The perspectivist literature argues for modeling the resulting label

distributions directly (Plank, 2022, Section 3); see also Pavlick and Kwiatkowski (2019).

Monotonicity and NLI. Monotonicity reasoning is a classic locus of difficulty in NLI. MacCartney (2009, Section 1.4.1) notes that downward-entailing behavior is not confined to obvious negation: “constructions which are not upward monotone are surprisingly widespread”, spanning quantifiers, conditional antecedents, superlatives, and open-class triggers across several parts of speech. The MED dataset targets monotonicity reasoning directly (Yanaka et al., 2019); we use it to anchor tagger error rates.

Operator semantics is itself contested. For several operators the monotonicity classification is a live semantic debate rather than a lookup: *only* has been analyzed as Strawson downward-entailing rather than simply downward-entailing (von Fintel, 1999), and *many* has received both cardinal and proportional readings (Rett, 2018, Section 2.1.2); for that ambiguity, Rett cites Partee (1989) among others. This is relevant to the audit in Section 5: part of the tagger-human disagreement we observe is irreducible for exactly this reason.

3 Method

Data. SNLI dev (9,986 items after excluding 14 items with four validation labels) and MNLI matched dev (10,000 items). MNLI mismatched dev (9,946 items after excluding 54 rows from 27 duplicated pair IDs) is analyzed as a preregistered secondary corpus. All corpora carry five labels per item. MED (5,382 premise-hypothesis pairs) is used only for tagger validation, and the ChaosNLI overlap (1,514 SNLI rows, of which 1,507 remain after the same four-label exclusion, and 1,599 MNLI matched items) only for the bridge analysis in Section 4.

Predictor. A rule-based monotonicity operator tagger (v0.3, frozen before this study under preregistration; no rule changes) labels each hypothesis upward, downward, non-monotone, or mixed by a surface trigger scan. The analysis contrast is binary: purely upward versus non-upward.

Outcome. A four-level agreement ordinal from the five validation labels: no majority $< 3/5 < 4/5 < 5/5$. The primary analysis includes the no-majority level; excluding it is sensitivity analysis A.

Statistics. Tie-corrected Cliff’s δ with two-sided Mann-Whitney U tests, percentile bootstrap CIs (10,000 resamples, seed 20260717), Holm correction over the two confirmatory contrasts (SNLI primary, MNLI matched primary), and a preregistered SESOI of $|\delta| = 0.10$. The registered prediction was $\delta < 0$ (non-upward items lower in agreement). The registration is a plan document written into our working repository on 2026-07-17, before the confirmatory analysis was run. Its timestamp is a version-control record under our own control rather than a third-party registry entry, so we report the date as a statement about our procedure and not as independently attested evidence; AUDIT.md in the reproduction repository sets out the registration timeline in those terms, and the full registration-versus-result concordance is in Appendix A.

Bridge to the earlier estimate. The originally planned correlation of the ordinal outcome with ChaosNLI entropy inside the overlap is undefined by arithmetic: ChaosNLI selected only items with a 3/5 modal label, so the agreement ordinal is constant (variance zero) on the entire overlap. We state this openly and replace the bridge with level-wise descriptive statistics (Section 4). Scale caveat: the earlier δ was computed on a 100-label entropy outcome and is not numerically commensurable with δ values on the four-level ordinal; we compare sign and magnitude class only.

4 Results

The registered prediction is rejected. Table 1 reports all seven contrasts. Every δ is positive: non-upward items sit very slightly higher, not lower, on the agreement ordinal. The SNLI primary contrast is not significant ($\delta = +0.032$, 95% CI $[-0.011, +0.075]$, Holm $p = 0.162$). The MNLI matched primary contrast is significant but with the sign opposite to the registration ($\delta = +0.045$, 95% CI $[+0.024, +0.066]$, Holm $p = 7.1 \times 10^{-5}$). The secondary mismatched corpus behaves like matched ($\delta = +0.059$). Dropping no-majority items (sensitivity A) and merging SNLI with matched (sensitivity B) change nothing material.

Effects are below the SESOI everywhere. All $|\delta| < 0.10$, and no bootstrap CI touches the boundary (Figure 1). Item-level discriminability is essentially absent: ordered logistic pseudo- R^2 is 0.000093 (SNLI) and 0.00084 (matched), and the

Contrast	Status	n non-up / up	δ	95% CI	p
SNLI primary	confirmatory	551 / 9,435	+0.032	$[-0.011, +0.075]$	0.162 (Holm)
MNLI matched primary	confirmatory	3,329 / 6,671	+0.045	$[+0.024, +0.066]$	7.1×10^{-5} (Holm)
MNLI mismatched	secondary	3,402 / 6,544	+0.059	$[+0.038, +0.079]$	4.1×10^{-8}
SNLI, no-majority dropped	sensitivity A	544 / 9,287	+0.030	$[-0.014, +0.072]$	0.193
Matched, no-majority dropped	sensitivity A	3,273 / 6,542	+0.044	$[+0.023, +0.065]$	5.3×10^{-5}
Mismatched, no-majority dropped	sensitivity A	3,351 / 6,427	+0.058	$[+0.036, +0.078]$	6.9×10^{-8}
SNLI + matched merged	sensitivity B	3,880 / 16,106	+0.045	$[+0.027, +0.063]$	1.0×10^{-6}

Table 1: All contrasts. Registered prediction: $\delta < 0$. SESOI: $|\delta| = 0.10$. No CI reaches the SESOI boundary. The earlier interval $[-0.332, -0.235]$ sits on a non-commensurable outcome scale, so comparison with it is one of sign and magnitude class only.

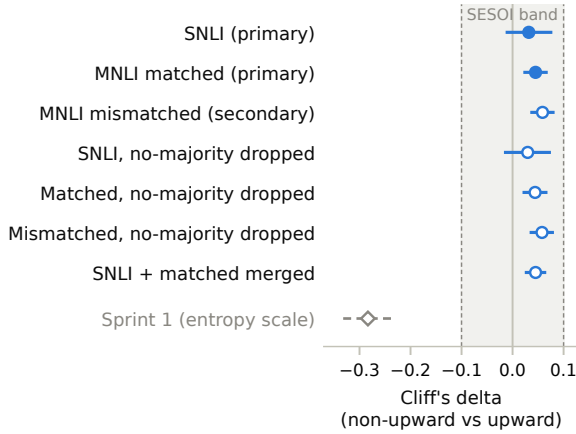


Figure 1: The seven contrasts of Table 1. Points are tie-corrected Cliff’s δ with percentile bootstrap 95% confidence intervals; the shaded band marks the preregistered SESOI at $|\delta| = 0.10$. The gray reference row is labeled *Sprint 1 (entropy scale)* after the earlier study’s internal name and reports that study’s estimate. It is computed on a 100-label entropy outcome that is not numerically commensurable with the four-level ordinal used here and is shown for sign and magnitude comparison only. Generated by analysis/fig_forest.py.

AUC for separating high-agreement (4/5, 5/5) from low-agreement items is at chance in both corpora (0.492 SNLI, 0.484 matched).

Outcome resolution does not explain the reversal. The earlier study read disagreement off 100 labels per item, while the ordinal used here has five levels, so one might ask whether the boundary failed to replicate only because the outcome lost resolution. Coarsening an outcome generally attenuates an association toward zero rather than reversing its sign, and every δ we observe is positive, so lost resolution cannot manufacture the direction we report. What resolution does bound is how precisely the two studies can be compared, which is why we restrict that comparison to sign and magnitude class. Subsampling experiments that vary

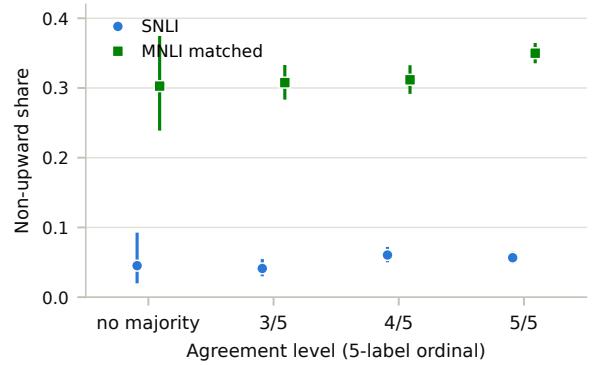


Figure 2: Share of non-upward hypotheses at each agreement level, with Wilson 95% confidence intervals, for SNLI dev and MNLI matched dev. The share does not decrease as agreement rises. Generated by analysis/fig_bridge_shares.py.

annotation count from one to one hundred labels per item on ChaosNLI make that dependence explicit (Kadasi and Singh, 2023).

Complexity confounders do not explain the reversal. As an auxiliary, non-confirmatory defense we fit proportional-odds models of the agreement ordinal on the binary monotonicity predictor, without (M0) and with (M1) length, parse depth, and genre covariates. In matched and mismatched the monotonicity coefficient keeps its sign and stays away from zero after adjustment (matched M1: -0.202 , $p = 2.4 \times 10^{-6}$; mismatched M1: -0.248 , $p = 7.3 \times 10^{-9}$; negative coefficients here correspond to the positive δ values above, because upward items sit lower in agreement). In SNLI the proportional-odds assumption fails a Brant test (Brant, 1990), and the multinomial substitute shows no significant category-wise monotonicity effect, which is consistent with the non-significant SNLI δ .

Level-wise structure runs against the registered direction. The share of non-upward hypotheses

does not decrease with agreement level. In matched it is highest among unanimous items: 0.303 (no majority), 0.308 (3/5), 0.312 (4/5), 0.350 (5/5). SNLI shows a flat low profile (0.045, 0.041, 0.060, 0.057). Figure 2 plots both corpora; Appendix B gives the arithmetic.

5 Validity of the Measurement

Tagger misclassification (Tier 2). We anchored flip rates in MED measurements of the tagger (symmetric error $542/5,382 = 0.101$; upward to non-upward $229/1,818 = 0.126$; non-upward to upward $313/3,564 = 0.088$; denominators recomputed from the released MED file, which differs by two items from the class counts reported in Yanaka et al., 2019; MED domain, not a dev-domain error claim) and simulated label flips over a 36-cell grid (three corpora, three scenarios, anchored plus fixed rates, 1,000 replications each). Misclassification shrinks δ toward zero rather than manufacturing it: at the MED-anchored symmetric rate the matched δ averages 0.0345 with significance retained in 94.9% of replications, and mismatched retains it in every replication. Retention drops only under the extreme fixed rates, most steeply for matched at symmetric 0.20 (0.594), where mismatched still holds at 0.933. Across all 36 cells the 97.5th percentile of simulated δ never exceeds 0.064, so the below-SESOI conclusion never flips (Appendix C).

Manual audit (Tier 3). We hand-tagged a fresh blind sample of 200 hypotheses (50 each from SNLI dev, MNLI matched dev, and the two ChaosNLI overlap strata) against a written codebook that fixes the judgment target: does the hypothesis contain a monotonicity trigger, judged by semantic definition rather than a word list. Four-class agreement with the tagger was 0.875 (Wilson 95% CI [0.822, 0.914]), binary agreement 0.905, Cohen’s κ 0.607 (four-class; both marginals are heavily skewed toward upward, so κ understates agreement and is read alongside the raw rates). An earlier round without a codebook produced agreement of 0.135. The gap is an instrument-specification result rather than an error: without a fixed judgment target, the human rater measured a different construct (premise-relative entailment direction rather than trigger presence). We report both rounds.

Three layers of residual disagreement. We read the 25 disagreements by hand (Appendix D) and

assigned each to an error layer. This layer coding is a qualitative judgment, not a script output; the counts below are approximate and are the one set of numbers in this paper that no script regenerates. They decompose into (a) about 10 tagger lexicon coverage gaps, (b) about 5 polysemy and construction failures, and (c) about 10 items where the operator’s monotonicity classification is itself semantically contested, including *only* (rated downward by the human 4 times and left upward 3 times within this sample; von Fintel, 1999) and *many* (Rett, 2018). Layers (a) and (b) are engineering-reducible; we judge layer (c) not to be, and it is itself direct evidence for treating disagreement as a measurement target. We deliberately did not patch the tagger in response to this audit, because reacting to the audit sample would circularize the validation.

6 Discussion

The leading interpretation is that the earlier boundary is a structure conditional on low-agreement selection. Inside a resource restricted to 3/5-split items, non-upward operators separated items by residual disagreement; in the unselected population, they do not, and the residual association even runs slightly the other way. Two implications follow. First, HLV structure claims estimated on selected re-annotation resources should carry their selection conditional explicitly; the ChaosNLI selection rule is strong enough to zero out the variance of the four-level agreement ordinal on the overlap, which is why the natural bridge analysis is undefined by arithmetic. Second, the disagreement decomposition shows that part of the tagger-human gap is the contested semantics of the operators themselves, which lexicon engineering is unlikely to resolve given the live debates documented in von Fintel (1999) and Rett (2018); measurement frames that treat disagreement as noise to be eliminated would misread exactly this layer. The preregistration, SESOI, and freeze discipline are what give the negative result its evidential value; we report it as registered.

7 Reproducibility

Every quantity this paper reports is printed in the paper itself, including the full appendix tables, so no claim requires external material. The reproduction repository (<https://github.com/oudeis01/nli-hlv-selection>) supplies

the code that regenerates those quantities. File paths named here are relative to that repository’s root.

All reported numbers regenerate from the analysis scripts, except the manual error layer counts in Section 5, which come from manual inspection of the items in Table 9. Every run that involves randomness fixes a seed and records it in that run’s output file. The repository README maps each table, figure, and statistic to the step that produces it, and gives the run order.

AUDIT.md is the audit trail. It records the registration timeline, what had already been observed at each registration point, every deviation from the registered plan, and every negative result. The repository is a curated snapshot, not a full development history, and AUDIT.md states what that costs.

Limitations

Manual tagging was performed by a single author, blind to tagger output but without an independent second rater; the reported κ is tagger-versus-human, not inter-human. The tagger is a surface trigger scanner with measured coverage gaps (layers (a) and (b) of Section 5). MED anchor rates are measured on MED sentences, not on the SNLI/MNLI dev domain. The four-level agreement ordinal is coarse, and all data are English. The earlier study and this replication use non-commensurable outcome scales, so the replication verdict rests on sign and magnitude class, not on interval overlap.

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A Registration-versus-Result Concordance

The preregistration is section 1 of the plan document written into the working repository and frozen

on 2026-07-17; the item numbers below refer to it, and the Registration column reproduces what each item registered, so the concordance can be read without consulting that document. Status values: “as registered” (followed without change), “prediction rejected” (procedure followed, registered direction failed), “documented deviation” (change made during the study, recorded in the analysis log before or at the point of change). Tables 2 and 3 give the full concordance.

B Bridge Details and Overlap Arithmetic

The original bridge design (correlating five-label agreement with ChaosNLI 100-label entropy on the overlap) is undefined by arithmetic: ChaosNLI selected items whose majority label carries exactly three votes, so the agreement ordinal has zero variance on the overlap. The overlap arithmetic confirms this exactly. All 1,599 ChaosNLI-M rows match MNLI matched dev items at level 3/5, and the 1,514 ChaosNLI-S rows decompose into 1,507 five-label items at 3/5 plus 7 four-label items at 3/4; the latter fall under the registered four-label exclusion, leaving 1,507 SNLI bridge items (Table 4).

The registered replacement has three components. Component 1 reports the spread of ChaosNLI 100-label entropy inside the constant-ordinal overlap, measuring the coarseness of the ordinal (Table 5): the SNLI overlap has mean entropy 0.800 (sd 0.332) and the matched overlap 1.072 (sd 0.267), so substantial graded variation survives within a single ordinal level. Component 2 reports level-wise non-upward shares with Wilson CIs (Table 6, plotted as Figure 2). Component 3 is this transparency statement itself.

C Full Tier 2 Grid

Table 7 lists all 36 cells: three corpora, three flip scenarios (symmetric, upward-to-non-upward only, reverse), and four rates per scenario arm (fixed 0.05, 0.10, 0.20 plus the MED-anchored rate for that scenario), with 1,000 replications per cell at seed 20260717. Reported per cell: mean simulated δ , the 2.5th and 97.5th percentiles, and significance retention. The maximum 97.5th percentile across all cells is 0.064, which is the basis for the claim in Section 5 that the below-SESOI conclusion does not flip under simulated misclassification.

D Tier 3 Codebook Summary and Disagreements

The codebook (v1, 2026-07-19) fixes the judgment target: whether the hypothesis sentence contains a monotonicity trigger, judged by semantic definition (does the expression license or block upward or downward substitution in its scope) rather than by membership in a word list. It defines the four-class label set (upward, downward, non-monotone, mixed), an existence-based decision rule, boundary-case guidance, and a circularity note: the codebook was written from semantic definitions, not from the tagger’s rule inventory, so agreement with the tagger is not built in.

Table 8 reports the full agreement metrics for the codebook round, including per-type disagreement counts. Table 9 lists all 25 disagreement items verbatim (cell, manual label, tagger label, detected triggers, hypothesis text), grouped by disagreement type. These 25 items are the basis for the three-layer decomposition in Section 5.

Registered item	Registration	Status
Outcome variable (1.1)	Four-level agreement ordinal, no-majority as lowest level, primary; three-level exclusion as sensitivity	as registered
Sample and exclusions (1.2)	SNLI dev minus 14 four-label items; MNLI matched in full; no other exclusions	as registered for confirmatory corpora; documented deviation for the secondary corpus (54 rows from 27 duplicated pair IDs in mismatched, excluded with logged rationale)
Predictor freeze (1.3)	Tagger v0.3 reused with no rule changes	as registered; the Tier 3 audit did not trigger modification, by explicit decision
Confirmatory hypothesis (1.3)	Non-upward group sits lower on the agreement ordinal	prediction rejected; all seven contrasts positive (Table 1)
Statistics (1.4)	Tie-corrected Cliff's δ , two-sided Mann-Whitney U, percentile bootstrap 10,000 draws, seed 20260717, Holm over the two confirmatory contrasts	as registered
Confound defense (1.4)	Proportional-odds model; multinomial substitute on assumption violation, violation reported	as registered; the violation occurred in SNLI and is reported in Section 4

Table 2: Registration-versus-result concordance, registered items 1.1 to 1.4.

Registered item	Registration	Status
Power and SESOI (1.5)	SESOI at absolute δ 0.10; report CI position against the boundary	as registered; every estimate is below the SESOI and no CI touches the boundary
Secondary and sensitivity analyses (1.6)	Mismatched pipeline, no-majority exclusion, merged corpus, outside the confirmatory family	as registered
Item-level ceiling (1.7)	Ordinal pseudo- R^2 and high-versus-low AUC with a noncommensurability caution	as registered (Section 4)
ChaosNLI bridge (1.8)	Three-component redefinition after the original correlation design was found undefined	as registered (Section 4 and Appendix B)
Tier 2 simulation (1.9)	Flip grid over fixed rates plus MED-anchored rates, 1,000 replications, seed 20260717	as registered (Appendix C)
Tier 3 manual audit (1.10)	200-item blind manual tagging, four cells of 50, single-author limitation stated	documented deviation: the first round lacked a written codebook and measured a different construct (agreement 0.135); a codebook was then written and a fresh 200-item sample re-tagged, with the original round excluded from the headline metric and both rounds reported (Section 5)

Table 3: Registration-versus-result concordance, registered items 1.5 to 1.10.

Metric	Value
<i>SNLI</i>	
chaos rows	1514
dev rows	10000
matched total	1514
matched 5-label	1507
matched 4-label	7
unmatched	0
agreement key 3/5	1507
agreement key 3/4	7
<i>MNLI matched</i>	
chaos rows	1599
dev rows	10000
matched total	1599
matched 5-label	1599
matched 4-label	0
unmatched	0
agreement key 3/5	1599

Table 4: ChaosNLI overlap arithmetic. Every matched item falls at a single agreement level, which is why the registered bridge correlation is undefined: the agreement ordinal has zero variance on the overlap.

Corpus	n	Mean	SD	Median	Q1	Q3	Min	Max
SNLI	1507	0.800	0.332	0.843	0.579	1.005	0.000	1.583
MNLI matched	1599	1.072	0.267	1.089	0.925	1.252	0.081	1.584

Table 5: ChaosNLI 100-label entropy inside the constant-ordinal overlap. Substantial graded variation survives within a single agreement level, which is the roughness the ordinal cannot express.

Level	n	Share	CI lo	CI hi
<i>SNLI</i>				
nomaj/5	155	0.045	0.022	0.090
3/5	1507	0.041	0.032	0.052
4/5	2845	0.060	0.052	0.070
5/5	5479	0.057	0.051	0.063
<i>MNLI matched</i>				
nomaj/5	185	0.303	0.241	0.372
3/5	1599	0.308	0.286	0.331
4/5	2457	0.312	0.294	0.330
5/5	5759	0.350	0.338	0.362

Table 6: Share of non-upward hypotheses at each agreement level, with Wilson 95% confidence intervals. The share does not decrease as agreement rises. Plotted as Figure 2.

Scenario	Rate	n valid	n degen.	Mean δ	2.5th pct	97.5th pct	Sig. ret.
<i>MNLI matched</i>							
directional_non_to_up	0.0500	1000	0	0.0439	0.0381	0.0492	1.000
directional_non_to_up	0.0878 (anchor)	1000	0	0.0430	0.0354	0.0510	1.000
directional_non_to_up	0.1000	1000	0	0.0430	0.0352	0.0514	1.000
directional_non_to_up	0.2000	1000	0	0.0409	0.0287	0.0529	0.990
directional_up_to_non	0.0500	1000	0	0.0407	0.0334	0.0483	1.000
directional_up_to_non	0.1000	1000	0	0.0373	0.0271	0.0471	0.995
directional_up_to_non	0.1260 (anchor)	1000	0	0.0362	0.0258	0.0463	0.992
directional_up_to_non	0.2000	1000	0	0.0320	0.0188	0.0454	0.909
symmetric	0.0500	1000	0	0.0396	0.0306	0.0501	1.000
symmetric	0.1000	1000	0	0.0345	0.0218	0.0470	0.946
symmetric	0.1007 (anchor)	1000	0	0.0345	0.0220	0.0465	0.949
symmetric	0.2000	1000	0	0.0249	0.0074	0.0398	0.594
<i>MNLI mismatched</i>							
directional_non_to_up	0.0500	1000	0	0.0573	0.0514	0.0631	1.000
directional_non_to_up	0.0878 (anchor)	1000	0	0.0562	0.0486	0.0635	1.000
directional_non_to_up	0.1000	1000	0	0.0557	0.0477	0.0633	1.000
directional_non_to_up	0.2000	1000	0	0.0531	0.0413	0.0639	1.000
directional_up_to_non	0.0500	1000	0	0.0537	0.0459	0.0607	1.000
directional_up_to_non	0.1000	1000	0	0.0490	0.0391	0.0588	1.000
directional_up_to_non	0.1260 (anchor)	1000	0	0.0473	0.0355	0.0581	1.000
directional_up_to_non	0.2000	1000	0	0.0423	0.0284	0.0550	0.998
symmetric	0.0500	1000	0	0.0518	0.0421	0.0607	1.000
symmetric	0.1000	1000	0	0.0454	0.0320	0.0581	1.000
symmetric	0.1007 (anchor)	1000	0	0.0450	0.0322	0.0577	1.000
symmetric	0.2000	1000	0	0.0330	0.0164	0.0499	0.933
<i>SNLI</i>							
directional_non_to_up	0.0500	1000	0	0.0317	0.0216	0.0413	0.000
directional_non_to_up	0.0878 (anchor)	1000	0	0.0317	0.0190	0.0454	0.001
directional_non_to_up	0.1000	1000	0	0.0319	0.0166	0.0465	0.000
directional_non_to_up	0.2000	1000	0	0.0317	0.0091	0.0547	0.016
directional_up_to_non	0.0500	1000	0	0.0172	-0.0065	0.0413	0.038
directional_up_to_non	0.1000	1000	0	0.0120	-0.0108	0.0342	0.037
directional_up_to_non	0.1260 (anchor)	1000	0	0.0103	-0.0120	0.0340	0.044
directional_up_to_non	0.2000	1000	0	0.0070	-0.0135	0.0270	0.025
symmetric	0.0500	1000	0	0.0161	-0.0080	0.0398	0.030
symmetric	0.1000	1000	0	0.0108	-0.0131	0.0353	0.041
symmetric	0.1007 (anchor)	1000	0	0.0108	-0.0147	0.0365	0.045
symmetric	0.2000	1000	0	0.0050	-0.0181	0.0270	0.026

Table 7: Tier 2 misclassification grid, all 36 cells. Seed 20260717, 1000 replications per cell. Rates marked (anchor) are the MED-anchored rate for that scenario. The maximum 97.5th percentile across all cells is 0.064, below the registered SESOI of $|\delta| = 0.10$.

Metric	Value	Metric	Value
n_items	200	disagreement type downward -> upward	7
four-class agreement rate	0.875	disagreement type upward -> non_monotone	6
four-class agreement n_agree	175	disagreement type downward -> non_monotone	3
four-class agreement n_total	200	disagreement type upward -> downward	3
four-class agreement wilson ci lo	0.822	disagreement type mixed -> downward	2
four-class agreement wilson ci hi	0.914	disagreement type mixed -> upward	2
binary agreement rate	0.905	disagreement type downward -> mixed	1
binary agreement n_agree	181	disagreement type non_monotone -> upward	1
binary agreement n_total	200	disagreement type mixed -> non_monotone	0
binary agreement wilson ci lo	0.856	disagreement type non_monotone -> downward	0
binary agreement wilson ci hi	0.938	disagreement type non_monotone -> mixed	0
four-class kappa	0.607	disagreement type upward -> mixed	0
binary kappa	0.688		

Table 8: Tier 3 agreement between the manual codebook labels and the tagger, with per-type disagreement counts. Directions read manual to tagger. Agreement is tagger versus human, not inter-human. The rows run down the left panel and continue in the right.

Item	Cell	Manual	Tagger	Triggers	Hypothesis
<i>downward -> upward (7 items)</i>					
7	chaosnli_s	downward	upward	(none detected)	50 people walked to the wrong subway hall.
34	chaosnli_m	downward	upward	(none detected)	It is better to plant when it is colder.
38	mnli_matched	downward	upward	(none detected)	It's impossible to have a plate hand-painted to your own design in Hong Kong.
47	mnli_matched	downward	upward	(none detected)	European members of NATO might consider the US's efforts to be less credible.
143	chaosnli_m	downward	upward	(none detected)	Was it Jane Eyre or not?
146	chaosnli_m	downward	upward	(none detected)	They would get upset whenever anyone would speak to them.
173	mnli_matched	downward	upward	(none detected)	We went to the office to see if there was anything we could rent.
<i>upward -> non_monotone (6 items)</i>					
21	mnli_matched	upward	non_monotone	only (non_monotone)	Zelon is the only student-run chapter of the American Civil Liberties Union in New York state.
58	chaosnli_m	upward	non_monotone	only (non_monotone)	Lister and Simpson were the only ones to use carbolic acid and chloroform for this purpose.
121	mnli_matched	upward	non_monotone	last (non_monotone)	Your mistress wrote letters last night.
128	chaosnli_m	upward	non_monotone	many (non_monotone)	Our higher-end stores have been suffering due to the recession, and many have shut down for lack of revenue.
184	chaosnli_s	upward	non_monotone	only (non_monotone)	There are only two people in the field.
188	mnli_matched	upward	non_monotone	last (non_monotone)	It was 37 degrees last night.
<i>downward -> non_monotone (3 items)</i>					
24	chaosnli_m	downward	non_monotone	only (non_monotone)	There were only a few villas the whole way along, until we reached a small village that seemed to be the end.
95	mnli_matched	downward	non_monotone	only (non_monotone)	Price hikes are only possible if there is more money in circulation.
119	mnli_matched	downward	non_monotone	smallest (non_monotone), only (non_monotone)	The library is the smallest estate in Jamaica, with only three books.
<i>upward -> downward (3 items)</i>					
3	chaosnli_s	upward	downward	each (downward)	Two dogs chase each other in the high grass.
66	chaosnli_m	upward	downward	n't (downward)	I wish you hadn't revealed your identity, that was a mistake.
181	snli_dev	upward	downward	little (downward)	A little girl is blowing the petals.
<i>mixed -> downward (2 items)</i>					
82	mnli_matched	mixed	downward	all (downward)	All of the homes in the hillside have been converted into art galleries and shops selling collectibles.
112	mnli_matched	mixed	downward	n't (downward)	I chose to become an actor, but I wasn't very good at it.
<i>mixed -> upward (2 items)</i>					
65	mnli_matched	mixed	upward	(none detected)	Everything can be found inside a shopping mall.
111	chaosnli_s	mixed	upward	(none detected)	The men are higher than the wall.
<i>downward -> mixed (1 items)</i>					
133	chaosnli_m	downward	mixed	only (non_monotone), barely (downward)	The tunnel of Eupalinos is only one foot in diameter, barely large enough for a child to squeeze through.
<i>non_monotone -> upward (1 items)</i>					
166	chaosnli_m	non_monotone	upward	(none detected)	Although it was unnecessary, some of the equipment was adjacent.

Table 9: All 25 Tier 3 disagreement items, grouped by disagreement type. Group headings read manual label to tagger label. These items are the basis for the three-layer decomposition in Section 5.